

PROTOCOL: Wearable IMU-Based Gait Monitoring System for Spatiotemporal and Dynamic Analysis

Smriti Dandin

April 2026

1 IMU Hardware Design

The wearable gait-analysis system comprises a custom-designed pelvic module and two shank-mounted IMU units, designed for portable, high-frequency data acquisition. The shank is selected due to its strong correlation with gait phase transitions, while the pelvic module enables centralized power management, data logging, and synchronization. The hardware framework was originally developed at the HCR Lab, IIT Gandhinagar, and was subsequently revised and re-engineered in this work to improve robustness, synchronization accuracy, and data reliability, including enhancements to hardware architecture and data acquisition pipelines. System components are detailed in Tables 1 and 2, and the complete setup is shown in Figure 1.

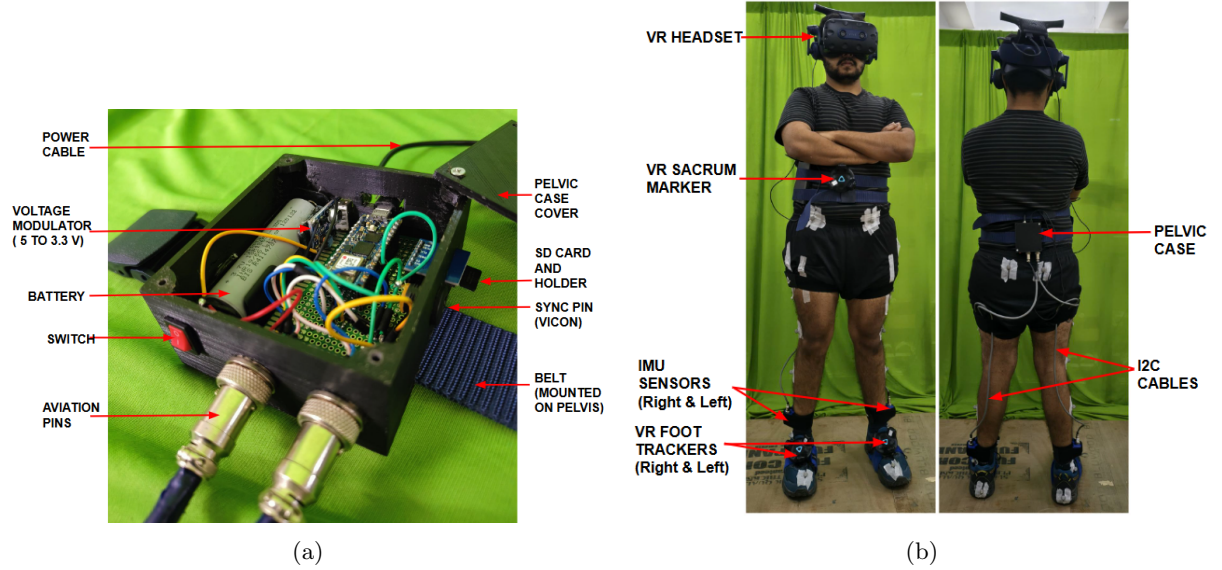


Figure 1: Wearable IMU and VR experimental setup: (a) Pelvic module, (b) Full-body IMU, VR and Vicon marker setup

1.1 Pelvic Module

Table 1: Components of the Pelvic Module

Component	Description
Pelvic Case (3D-printed)	3D-printed enclosure worn on the pelvis with a hinged lid for access to internal electronics.
Microcontroller (Arduino Nano RP2040)	Handles IMU data acquisition, timestamping, SD-card logging, and Vicon synchronization.
Custom PCB	Provides reliable I2C routing, power distribution, and pin connectivity.
Voltage Regulator (5V to 3.3V)	Ensures stable power delivery to the RP2040 and IMUs.

Component	Description
3.7V Li-ion Battery	Rechargeable power source supporting several hours of operation.
Power Switch	External toggle switch for system on/off control.
MicroSD Module	High-speed logging of sensor data, quaternions, timestamps, and gait events.
USB–MicroUSB Port	Used for powering/programming the microcontroller during setup.
Vicon Sync Pin	Receives Vicon trigger pulse for perfect IMU–Vicon time alignment.
Aviation Connectors	Rugged connectors linking pelvic module to shank IMUs via shielded cables.

1.2 Shank-Mounted IMU Units

Table 2: Components of the Shank-Mounted IMU Units

Component	Description
IMU Sensor (ICM-20948)	9-axis sensor providing accelerometer, gyroscope, and magnetometer data. Right IMU uses I2C address <code>0x69</code> , left uses <code>0x68</code> .
Mounting Straps	Elastic straps used to securely mount each IMU on the medial shank with consistent anatomical alignment.
4-Core I2C Cable	Carries SDA, SCL, GND, and 3.3V between the pelvic module and IMUs.

2 Virtual Reality Experimental Setup

To investigate locomotor stability under controlled perturbations, the wearable IMU system was integrated with an immersive virtual reality (VR) platform. VR enables the safe introduction of controlled perturbations while maintaining precise regulation of visual stimuli and walking conditions. In this study, a structured virtual environment was designed to evaluate adaptive gait responses during continuous locomotion. The VR system consisted of a head-mounted display (HMD) and motion trackers used to capture participant movement within the virtual environment. Two trackers were attached to the feet, and an additional tracker was placed at the sacrum to approximate whole-body motion and serve as a proxy for the center of mass. These trackers provided real-time positional data for reconstructing motion in the virtual space, while the HMD delivered immersive visual feedback.

A virtual bridge environment (Figure 4) was designed to simulate a constrained walking pathway within a physical space of approximately 6 meters. Directional cues were incorporated to guide turning at the ends of the walkway, enabling continuous movement within the limited space. Two types of walking trials were conducted: **baseline trials** (walking without perturbations) and **perturbation trials** (walking with visually guided cues), with each lap typically containing one perturbation event. During perturbation trials, a target box appeared during selected strides, requiring participants to place a specific foot based on a visual cue: a **white box** indicated the **left foot**, while a **green box** indicated the **right foot**. Post-processing was performed to remove turning phases between laps and eliminate confounding effects. Trials with incorrect responses, such as stepping with the wrong foot or missing the target box, were excluded from analysis.

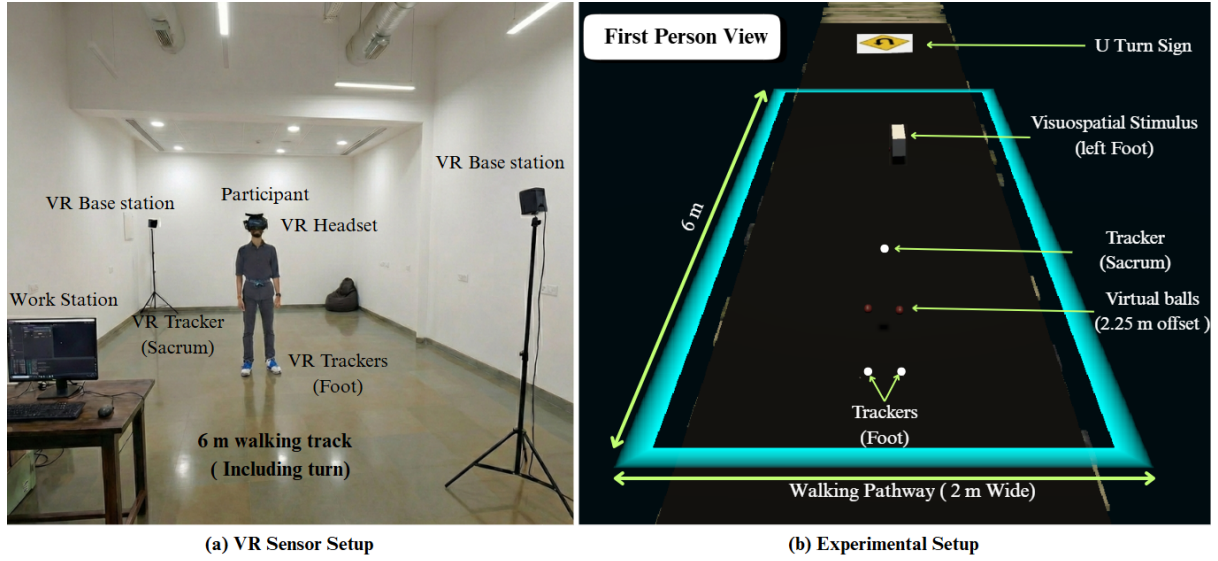


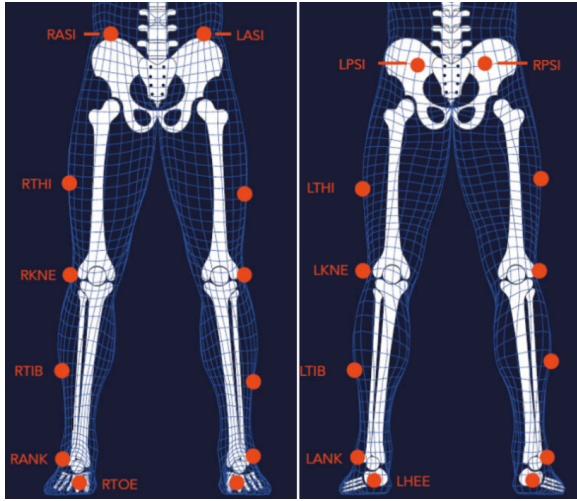
Figure 2: VR setup and the immersive environment configuration used for perturbation-based walking experiments.

2.1 Vicon Nexus and Plug-in-Gait Lower Body Model

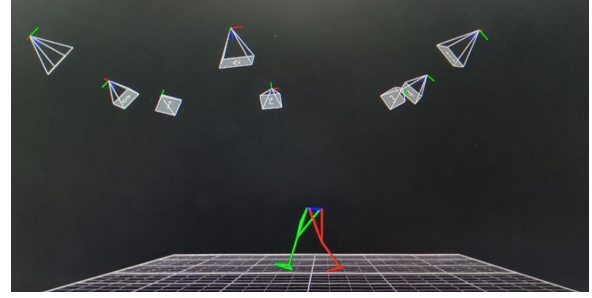
The Vicon motion capture system is a widely used biomechanical measurement platform that employs multiple infrared cameras to track reflective markers with sub-millimeter accuracy. It provides high-precision three-dimensional kinematic data and is considered the gold standard for gait analysis. In this study, all recordings were processed using Vicon Nexus software, which performs marker labeling, three-dimensional reconstruction, and trajectory smoothing. Prior to each trial, the system was calibrated by defining the capture volume and aligning the cameras using a Vicon calibration wand. Lower-limb kinematics were captured using the Plug-In Gait (PiG) model, which utilizes 16 reflective markers placed on key anatomical landmarks of the pelvis, thigh, knee, shank, ankle, heel, and toe. These markers enabled accurate detection of gait events, including heel strike and toe-off, allowing precise computation of spatiotemporal gait parameters for validation of the IMU-based system.

Table 3: Plug-in Gait lower-body marker configuration

Region	Right Marker	Left Marker	Anatomical Location
Pelvis	RASI	LASI	Anterior Superior Iliac Spine (ASIS)
Pelvis	RPSI	LPSI	Posterior Superior Iliac Spine (PSIS)
Thigh	RTHI	LTHI	Mid-thigh, lateral surface
Knee	RKNE	LKNE	Lateral femoral epicondyle
Shank	RTIB	LTIB	Mid-shank, lateral surface
Ankle	RANK	LANK	Lateral malleolus
Heel	RHEE	LHEE	Calcaneus (heel)
Toe	RTOE	LTOE	Second metatarsal head (toe)



(a) Plug-in Gait marker placement (front and back)



(b) Vicon Nexus interface

Figure 3: Vicon system setup

3 Experimental Protocol

Controlled gait experiments were conducted to evaluate the proposed framework under both overground and treadmill walking conditions. Data were recorded using the wearable IMU system, Vicon motion capture system, and, where applicable, the VR platform. The protocol included participant preparation, system setup and synchronization, walking trials, and post-processing of the recorded datasets.

3.1 Participant Preparation and Instrumentation

- Participants were instructed to wear tight-fitting clothing to ensure accurate reflective marker tracking and to minimize soft tissue artefacts.
- Before the experiment, all required devices including IMU sensors, VR trackers, VR headset, and controllers were fully charged and checked for proper functioning.
- A standard 16-marker lower-body Plug-in Gait model was applied according to the Vicon protocol.
- Reflective markers were placed on the required anatomical landmarks for lower-body gait analysis.
- After marker placement, all Vicon cameras and IR blasters were switched on and checked to ensure full marker visibility inside the capture volume.
- The pelvic IMU case was secured properly around the pelvis.
- IMU sensors were attached to the right and left shanks and firmly fastened using Velcro straps to prevent movement artefacts during walking.
- For overground VR trials, Vive trackers were attached to:
 - Sacrum
 - Right foot
 - Left foot
- All trackers were secured carefully to prevent displacement during gait.
- Finally, the participant wore the VR head-mounted display (HMD), and comfort, fit, and visibility were checked before proceeding.

3.2 System Setup, Calibration, and Synchronization

- The VR headset and Vive trackers were switched on first.
- Vive Wireless was launched, which automatically initialized SteamVR and connected the headset.
- The virtual walking environment was opened using Unity Hub.
- Inside Unity:
 - Right foot tracker was mapped
 - Left foot tracker was mapped
 - Sacral tracker was mapped
- The VR headset was re-centered to ensure proper alignment of the participant within the virtual environment.
- Both red virtual foot markers (red balls) were checked for visibility to confirm correct tracker mapping.
- Proper functioning of the virtual environment was verified before data collection.
- Vicon Nexus was then opened for motion capture setup.
- The capture volume was calibrated using the Vicon calibration wand.
- Camera origin was set carefully, ensuring that the participant and VR headset remained fully inside the calibrated capture space.
- A static live trial was first recorded to verify:
 - All markers were visible
 - Markers were properly labelled
 - Tracking quality was complete and stable
- Two separate systems (PCs/devices) were used:
 - One for Vicon Nexus
 - One for VR and Unity
- Vicon functioned as the master clock for synchronization.
- Unity functioned as the slave system and waited for the trigger signal from Vicon.
- Synchronization was established as follows:
 - IMU system using a physical sync cable
 - Unity/VR system using network-based trigger signals from Vicon
- In Vicon Nexus:
 - Auto Capture mode was enabled
 - “Send via Network” was selected
 - System was locked and armed
- This allowed Unity to listen for the Vicon trigger signal and automatically begin recording when Vicon capture started.

3.3 Data Acquisition and Walking Trials

- Before each experimental trial, a 10-second static calibration trial was recorded with the participant standing still.
- This initialization phase was used to:
 - Capture baseline sensor signals
 - Reduce measurement noise
 - Stabilize IMU readings
- The sync cable was connected to the IMU system.
- The IMU system was switched on and allowed a 10-second initialization/calibration period.
- Unity was started and allowed to listen for the synchronization signal.
- Vicon recording was then started using armed Auto Capture mode.
- After Vicon started recording, a 2–3 second buffer period was provided.
- The sync cable was then removed, and the participant was instructed to begin walking.
- **Overground Walking – Baseline Trial**
 - Baseline mode was selected in Unity
 - No perturbation was applied
 - Participant walked continuously for 20 laps inside the VR environment
 - Participant was instructed to follow the turn symbol displayed in VR
 - When the turn symbol appeared, the participant turned and continued walking
 - If the turn symbol was visible immediately at trial start, the participant first completed a half lap before continuing normal walking
- After completion:
 - IMU recording was stopped
 - VR recording was stopped
 - Vicon recording was stopped
 - All files were saved properly
- **Overground Walking – Perturbation Trial**
 - Same procedure was repeated
 - Perturbation mode (Stroop condition) was selected in Unity
 - Synchronization steps were repeated exactly
 - Data caching was enabled only when actual recording was required
- All files were named systematically for synchronization, for example:
 - SUB1_VR_Baseline
 - SUB1_IMU
 - SUB1_VICON_Lap1
- **Treadmill Walking Trial**
 - VR system was not used
 - Only IMU and Vicon systems were active
 - Same sync cable and initialization procedure was followed
 - Participant walked on the treadmill at a constant comfortable speed of approximately 3–4 km/h
 - Walking duration was approximately 3–5 minutes per trial

3.4 Data Recording and Post-Processing

All experimental data were stored subject-wise inside the **SmritiDandin_MSThesis_HCRLab_IITGn** folder, with separate folders for codes and all 10 experiment processed files for IMU, Vicon, and VR recordings in CSV format. Files were maintained using consistent naming conventions to ensure proper synchronization across all three systems during post-processing.

3.4.1 IMU Data Processing

- Raw IMU data were recorded as complete CSV files containing both the initialization phase and the walking phase.
- The MATLAB script

`Python Codes/splitCSV`

was used to divide each raw IMU file into two standardized outputs: `imu_initialization.csv` and `imu_detection.csv`.

- The `imu_detection.csv` file was then imported into

`Final Codes_HCR_Smriti/PythonCodes/Gait Event Detection_Final`

for detection of key gait events such as heel strike and toe-off.

3.4.2 Temporal Gait Validation (IMU vs Vicon)

- Validation of temporal gait parameters such as stride duration, step time, and cadence between IMU and Vicon was performed using

`Final Codes_HCR_Smriti/PythonCodes/Gait Event Vicon Validation/Temporal Validation`

- The required inputs were `imu_detection.csv` and the corresponding Vicon CSV files.

3.4.3 Treadmill Validation

- Validation of temporal and spatial gait parameters during treadmill walking trials (without VR) was performed separately using

`Final Codes_HCR_Smriti/PythonCodes/Treadmill_validation code`

3.4.4 Vicon Stride Length Computation

- Stride length from the Vicon reference system was computed using

`Final Codes_HCR_Smriti/PythonCodes/Vicon SL`

- This served as the laboratory gold-standard reference for spatial gait validation.

3.4.5 VR Stride Length and Dynamic Stability Computation

- Raw VR files were first processed using the MATLAB script

`Matlab Codes/Wrong Lap Removal`

to remove incorrect laps and generate a filtered VR output.

- The filtered VR file was then imported into

`VR_SL Code`

to compute stride lengths and generate the cleaned stride length output file: `vr_Filtered_StrideLength_cleanStri`

- Upload the filtered file was further imported into

`Master3Code_VR_MOS`

to compute dynamic stability measures and generate the output file `vr_Filtered_MOS_at_HS`, representing Margin of Stability (MoS) at heel strike.

3.4.6 IMU Stride Length Computation and Three-System Comparison

- Stride length computation and synchronized comparison across IMU, VR, and Vicon systems were performed using

Final Codes_HCR_Smriti/PythonCodes/

- Master3Code_VR_IMU_VICON_BASELINE
- Master3Code_VR_IMU_VICON_PERTURBATION

- Required inputs included the raw IMU CSV file, `imu_detection.csv`, `vr_Filtered_StrideLength_cleanStrides.csv`, all Vicon lap files from the Trials folder, and `vr_Filtered_MoS_at_HS`.
- The outputs provided synchronized stride length comparison across IMU, VR, and Vicon systems for both baseline and perturbation trials.

3.4.7 Correlation Analysis of Dynamic and Spatial Parameters

- Correlation analysis between dynamic stability measures and spatial gait parameters was performed using

Final Codes_HCR_Smriti/PythonCodes/

- MasterCode_Baseline_Correlation_DynamicSpatial
- MasterCode_Perturbation_Correlation_DynamicSpatial

- Required inputs included raw IMU files, `imu_initialization.csv`, `imu_detection.csv`, VR files, `vr_Filtered_StrideLength_cleanStrides.csv`, and `vr_Filtered_MoS_at_HS`.
- This analysis established relationships between spatial gait parameters such as stride length and dynamic stability parameters such as Margin of Stability (MoS) and Lyapunov Exponent (LyE). The analysis was performed separately for baseline and perturbation walking trials.

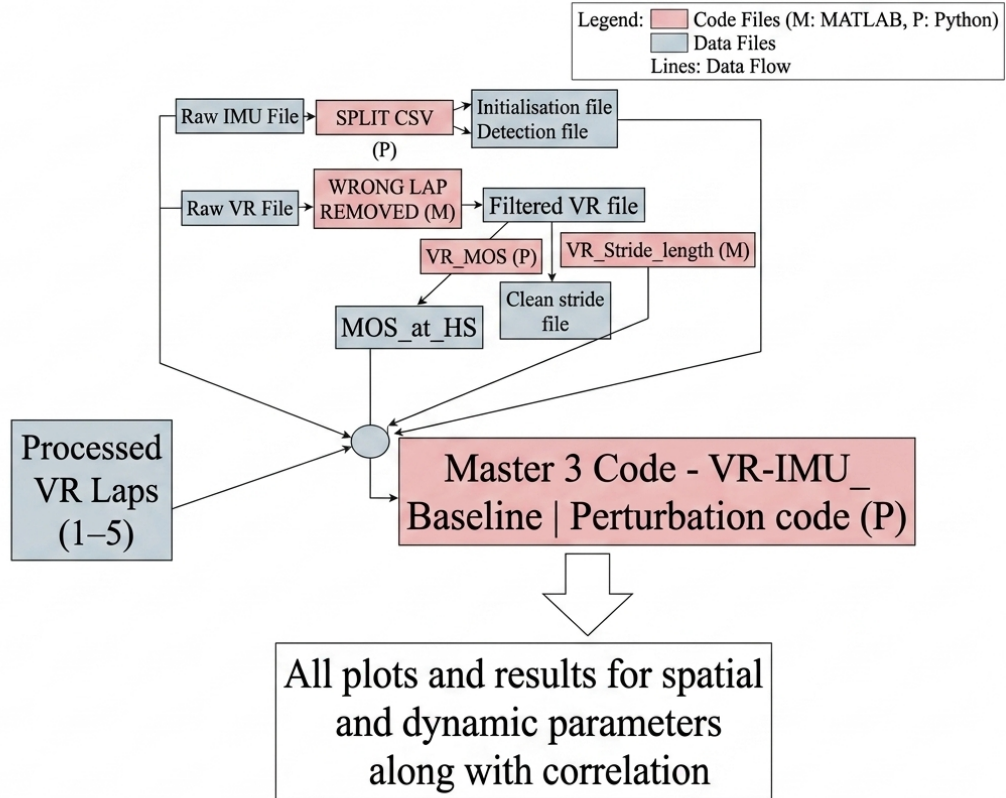


Figure 4: Flowchart of postprocessing

3.5 Notes

- Make sure all ethical guidelines are followed and that the participant feels safe and comfortable throughout the experiment.
- While using the real-time Arduino code, ensure that `UseViconSync` is set to `false` if Vicon is not being used.
- In the Arduino code folder under **Extra Codes**, I have included all the intermediate codes I used step by step to understand and build the final code.
- I had earlier worked with quaternions (extracted from the Madgwick filter) and studied the rotation matrix between local and global frames in detail. Refer to those codes whenever needed.
- Always verify functionality through serial printing before conducting the experiment. For example, check whether the IMU and VR setup batteries are fully charged, and confirm messages such as *“Right/Left shank IMU initialized successfully!”* appear correctly. If there is any issue, inspect the hardware wiring and soldering on both ends of the I2C cables.
- Always remember to insert the SD card before conducting the experiment, and save the files immediately afterward to avoid data loss or crashes.
- Ensure proper synchronization during the trial and save all three files for each trial.